to remain until dry. Drying takes place by a combination of evaporation and gravity drainage through the sand.

Sludge Disposal

The final destination of treated sewage sludge usually is the land. Dewatered sludge can be buried underground in a sanitary landfill. It also may be spread on agricultural land in order to make use of its value as a soil conditioner and fertilizer. Since sludge may contain toxic industrial chemicals, it is not spread on land where crops are grown for human consumption.

When a suitable site for land disposal is not available, as in urban areas, sludge may be incinerated. Incineration completely evaporates the moisture and converts the organic solids into inert ash. The ash must be disposed of, but the reduced volume makes disposal more economical. Air pollution control is a very important consideration when sewage sludge is incinerated. Appropriate air-cleaning devices such as scrubbers and filters must be used.

Benefits of Sewage Treatment

- Water recycling has proven to be effective and successful in creating a new and reliable water supply, while not compromising on public health.
- Water recycling can help us find ways to decrease the diversion of water from sensitive ecosystems
- Water users can supplement their demands by using recycled water.
- Decreases wastewater discharges
- Reduces and prevents water pollution
- Recycled water can be further used in Thermal power plant (for cooling), Municipal use, Irrigation and Agricultural use.

Infobits

There is a huge business opportunity in finding ways to use these waste dumps for productive purposes - energy, organic fertilizer. This requires methods of dealing with **old waste** that has been accumulating over the years as opposed to new/fresh waste.

9.7 Composting

Compost is organic matter that has been decomposed and recycled as a fertilizer and soil amendment. It is a mass of rotted organic matter made from waste. Example: garbage, paper, sugarcane trash, paddy straw, aquatic weeds, other agricultural waste.

Composting is a natural process in which aerobic and anaerobic microorganisms decomposes organic matter into valuable manure called as compost. The primary objective of composting is to convert an unstable material into stable end product (Figure 9.11).

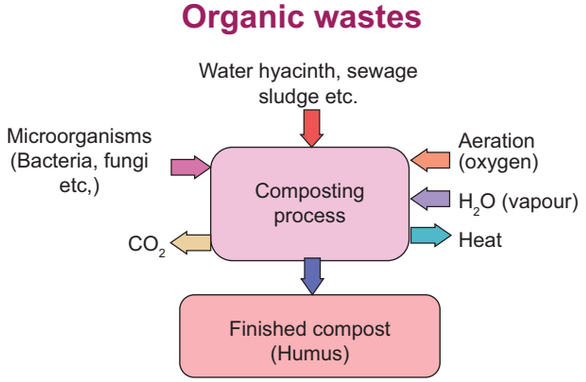


Figure 9.11: Composting process

The humification of organic material occurs in three stages

1. **Mesophilic stage** - Mesophilic is the initial stage of decomposition, lasting

TN_GOVT_XI_Micro_Biology_CH09.indd 140 16-02-2019 10.02.27 AM